



Review

A meta-analysis of sequences in theory-of-mind understandings: Theory of mind scale findings across different cultural contexts

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ABSTRACT

In 2004, Wellman and Liu demonstrated that theory of mind (ToM) – the awareness of how mental states such as desires, beliefs, and intentions govern actions – develops via a progression of understandings, and pioneered a 5-item ToM Scale to measure those sequential developments. This ToM Scale has now been translated into many non-English languages and used worldwide. We conducted a meta-analysis to quantitatively summarize the now-abundant ToM Scale findings (across 91 studies and 10,321 2- to 10-year-old children) and describe how children worldwide progress through a sequence of ToM understandings in early development. Results showed that the ToM Scale successfully captures a robust developmental progression of ToM understandings. It also captures universality and cultural-specificity in ToM development: while ToM universally develops in a progressive sequence where initial insights lead to later ones, children from individualistic and collectivist countries progress through two slightly different sequences shaped by cultural learning.

Introduction

Theory of mind (ToM) – the awareness of how mental states such as beliefs, desires, and intentions govern the behavior of self and others – is a cornerstone of social intelligence and satisfying social interaction, and hence social-emotional life (Wellman, 2014; Yu & Wellman, 2023). ToM develops rapidly during the preschool period and this development is classically demonstrated via a single ToM milestone, understanding false beliefs – mental states that decidedly diverge from the real state of affairs. Typically, children worldwide develop from consistently failing to consistently passing false belief tasks between the ages of 3 and 7 years (Wellman et al., 2001). However, focusing on a single task or achievement is limited, and research accumulated over the past 20 years has shown that ToM develops over many years in an extended sequence of understandings.

In 2004, Wellman and Liu pioneered a now often-used method for examining a key set of these sequential ToM understandings (Wellman & Liu, 2004) – a ToM Scale. The ToM scale consists of five child-friendly behavioral tasks with similar procedures, language, and format, tapping different aspects of understanding persons' mental states. It assesses children's understanding that people's desires can differ (diverse desires; DD), that people's beliefs can differ (diverse beliefs; DB), that people can be knowledgeable or ignorant (knowledge access; KA), that they can have mistaken ideas (false beliefs; FB), and that people may feel something inside that is not displayed outside on their face or in their actions (hidden emotion; HE).

That initial study (Wellman & Liu, 2004) demonstrated that the patterns of performance on the five tasks not only order themselves

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in a developmentally plausible manner, as shown in Table 1 (with different percentages of children passing the task), but analytically, they also define a Guttman scaling sequence (Green, 1956), where it is rare for a child to pass a later step without passing all the preceding ones. Longitudinal research shows that this cross-sectional sequence also reflects the sequence of understandings individual children go through as they grow older (e.g., Peterson & Wellman, 2019; Wellman et al., 2011). Helpfully, the scale has been extended to include later-developing ToM understandings such as understanding non-literal language (e.g., sarcasm), understanding interpretive mental stances, and understandings of conscious versus unconscious states of mind (e.g., Peterson et al., 2012). Importantly, because the ToM scale encompasses a series of conceptual sequences that children typically follow in their development, it emphasizes and advocates for a broader developmental construal of ToM development and enables a more precise exploration of key questions, such as how and which aspects of ToM understanding might be universal across different contexts.

Since 2004, this ToM Scale has been translated into more than a dozen non-English languages and used worldwide, with the original article having more than 2,800 citations (as of 2024, according to Google Scholar). Some reviews summarizing some of this data in qualitative, narrative forms have provided helpful insights about the sequential development of these ToM understandings (e.g., Beaudoin et al., 2020; Fu et al., 2023). However, they fall short of providing a systematic, quantitative integration of the findings as would be available in a meta-analysis of the ToM Scale data. Such a meta-analysis could resolve discrepancies apparent when only a handful of studies are selectively, qualitatively compared and could provide a detailed summary and quantitative analyses of the specificity and universality of the ToM-Scale findings, including revealing the extent to which ToM sequences are or are not common across different cultural contexts. In this article, we provide a comprehensive meta-analysis on the ToM Scale addressing these issues.

Conducting a meta-analysis is an important endeavor. It allows researchers to address issues of replicability and failure to replicate, an important topic in contemporary psychological science (Nosek et al., 2022). It also helps address complaints about the plethora of small-sample, underpowered research studies, including studies with children (Davis-Kean & Ellis, 2019). Indeed, meta-analysis is often offered as a crucial and practical way to address these issues because it aggregates many, often small-sample studies into a larger dataset encompassing hundreds, even thousands, of participants (Hedges & Schauer, 2019). Additionally, a meta-analysis, depending on the studies available, can also help overcome the often-lamented predominance of research with WEIRD participants (those from western, educated, industrialized, rich, and democratic contexts; Henrich et al., 2010) by including data from other, less represented populations. Overall, these meta-analytic advantages not only help address methodological concerns but also advance substantive research goals by achieving a more valid picture of childhood development worldwide.

Such a meta-analysis of the ToM Scale is now feasible, because the number of studies using this “standard” ToM Scale is now large and they are widespread geographically. The studies are not only sufficient for an informative meta-analysis, but also so large and widespread that considering all the studies one by one is difficult. The populations sampled worldwide include large samples of English-speaking children (e.g., from the US, the UK, Canada, and Australia), European children (e.g., from Germany, France, and Switzerland), and Asian children (e.g., from China, Indonesia, and Singapore) from both large countries (e.g., the US, Brazil, China) and smaller ones (e.g., New Zealand, Singapore, Vanuatu). Given this, a meta-analysis of the ToM Scale could contribute to our understanding of ToM development worldwide and more specifically, address questions of specificity and universality that are almost never possible in a single study.

Analytically, it is helpful that almost all ToM Scale studies report raw percentages of children passing the five tasks (as in Table 1), allowing straightforward aggregation of an easy-to-understand statistic. And most report Guttman statistics – especially Guttman’s Coefficient of Reproducibility (Green, 1956) – which quantify the extent to which individuals adhere to the overall sequence, allowing direct comparison between studies. A meta-analysis of such data is straightforward; many of the statistical problems that arise for other types of meta-analyses, which require the re-computation and collapsing of various derived inferential statistics (e.g., *F* tests, *t* tests; Glass et al., (1981), can be avoided given this type of comparable data across studies.

Our principal goals are descriptive: We aim to quantitatively summarize the extensive findings from the ToM Scale (across 91 studies and 10,321 2- to 10-year-old children; see Methods) and examine how children across the world progress, similarly and/or differently, through a sequence of ToM understandings in their development of insights into persons and minds. However, two specific predictions, guided by previous findings and several qualitative reviews, emerge in advance. The first concerns ToM developmental timetables as indexed by age: The five milestones in the ToM Scale focus predominantly on preschool understandings, at least for typically-

Table 1
Summary of the Tasks and Children’s Performance from Wellman & Liu (2004).

Task	Proportion correct	Description
Diverse desires (DD)	95 %	Child judges that two persons (the child vs. someone else) can have different desires about the same object.
Diverse beliefs (DB)	84 %	Child judges that two persons (the child vs. someone else) can have different beliefs about the same object (when the child does not know which belief is true or false).
Knowledge access (KA)	73 %	Child sees what is in a container and judges whether another person who has never seen inside it will know what the container holds.
False beliefs (FB)	59 %	Child sees a distinctive container (e.g., a crayon box) and looks inside to find what it has (e.g., small candles), then judges what someone who has never seen into the container will think it holds.
Hidden emotion (HE)	32 %	Child judges that a person can feel one thing but display a different emotion on his/her face.

Note. The table is Table 3 in the original paper adjusted only in its formatting and wordings with authors’ approval.

developing children. Thus, we expect that most children worldwide will have developed all or most of these understandings by 6 or 7 years of age. The second prediction concerns universality and specificity in ToM development: We expect the sequence of acquisition of these understandings to be largely universal but with some constrained variation. Many studies of North-American and European children growing up in individualistic cultures have found children's performance to conform to the sequence depicted in Table 1 (DD-DB-KA-FB-HE), including that understanding DB precedes understanding KA. Children growing up in collectivist cultures (such as China or Iran) have evidenced a slightly different sequence where understanding KA precedes understanding DB (i.e., DD-KA-DB-FB-HE). Because, to the best of our knowledge, no other dominant sequences have been reported, we predict that meta-analysis will confirm the presence of only these two sequences (and their associated cultural contexts) – a pattern of essentially universal development coupled with limited and specific cultural variation.

Methods

Literature search

We searched for and reviewed all available published child studies using Wellman and Liu's ToM Scale. Studies were identified for inclusion by conducting searches in PsycINFO, PsycArticles, Child Development & Adolescent Studies, Education Abstracts, Family & Society Studies Worldwide, and Social Sciences Abstracts for two keywords: "theory of mind," and "theory-of-mind." We applied the keywords to search each database for articles published between January 2004 and March 2024. We restricted the age to childhood (birth to twelve years). We further directly identified published articles and the "grey literature" (e.g., dissertations, preprints, government reports, or conference proceedings) that cited the original Wellman and Liu paper. These search procedures returned a total of 5829 studies after the removal of duplicate studies.

The 5829 studies were then reviewed for the following inclusion criteria. First, the study had to report original and empirical research. Case studies, review papers, meta-analyses, and opinion papers were excluded. Second, the study had to use the focal ToM Scale. Third, the study had to include typically-developing children aged younger than 12 years. Studies with only atypically-developing children such as children with deafness or developmental difficulties were excluded. For studies with both typically and atypically-developing children we extracted data for only the typically developing children. Fourth, the study had to be written in English.

Fig. 1 shows a PRISMA diagram of the steps involved in the search and exclusion process. From reviewing all Abstracts, it was possible to exclude 5364 studies, leaving 465 studies for full article review. Reviewing these 465 articles in detail resulted in excluding 319 that failed to meet our search criteria. Inter-rater reliability for these search decisions was high with a consensus rate of 96 %. When questions arose about inclusion criteria, all reviewers discussed the study until a consensus about its inclusion was reached. These processes reduced the database to 146 potential studies. Of these 146 potential studies, some studies reported ToM Scale results in a clear and extractable format; some studies, however, did not (e.g., they merely used a total ToM Scale score as a control variable in other analyses). We contacted the authors of all these 146 potential studies to ensure we had the most complete and accurate information for each study.

Our final database includes 91 studies.¹ All these studies reported the results of the ToM Scale in either (1) the raw percentages of children passing the tasks along with their age (e.g., reported the percentages correct on diverse-desire, diverse-belief, and knowledge-access tasks for a group of 4-year-old children), and/or (2) the Guttman Coefficient of Reproducibility (Green, 1956) along with the corresponding sequences (e.g., in a group of 4-year-old children, reported a Guttman Coefficient of Reproducibility of 0.96 for a DD-DB-KA-FB-HE sequence). Both formats provide sufficient useful information to examine the ToM-Scale pattern and quantify the extent to which a group of children adhered to a certain sequence. See below for more details on data coding and analysis.

The included 91 studies encompass a total of 10,321 children with 141 data points (i.e., some studies include several groups of children thus contributing more than one data point). Each data point represents the average performance of a group of participants. For example, one data point may indicate that in a group of 75 US children aged 4.5 years, 95 %, 84 %, 73 %, 59 %, and 32 % were correct on the DD, DB, KA, FB, and HE tasks respectively, with a Guttman Coefficient of Reproducibility of 0.96 following a DD-DB-KA-FB-HE sequence.

Data coding and analysis

Studies that met the inclusion criteria were coded for several elements. Two reviewers coded each study first, and a third reviewer checked these coding decisions. When questions or initial disagreements arose about the coding, all reviewers discussed them until consensus was reached.

First, the items of the ToM Scale and the corresponding percentages were coded for each data point. Most studies reported all five items (i.e., DD, DB, KA, FB, and HE), although some only reported a subset of the items. The average age was also extracted for each data point.

Second, the reported Guttman Coefficient of Reproducibility (Green, 1956), along with the corresponding sequence used to compute that index, was extracted for each data point. This coefficient is a statistical measure of how well the item difficulties

¹ No difference was found between included and non-included studies in terms of culture ($\chi^2(1) = 0.31, p = 0.58$) or age ($t(62.174) = 0.28, p = 0.78$).

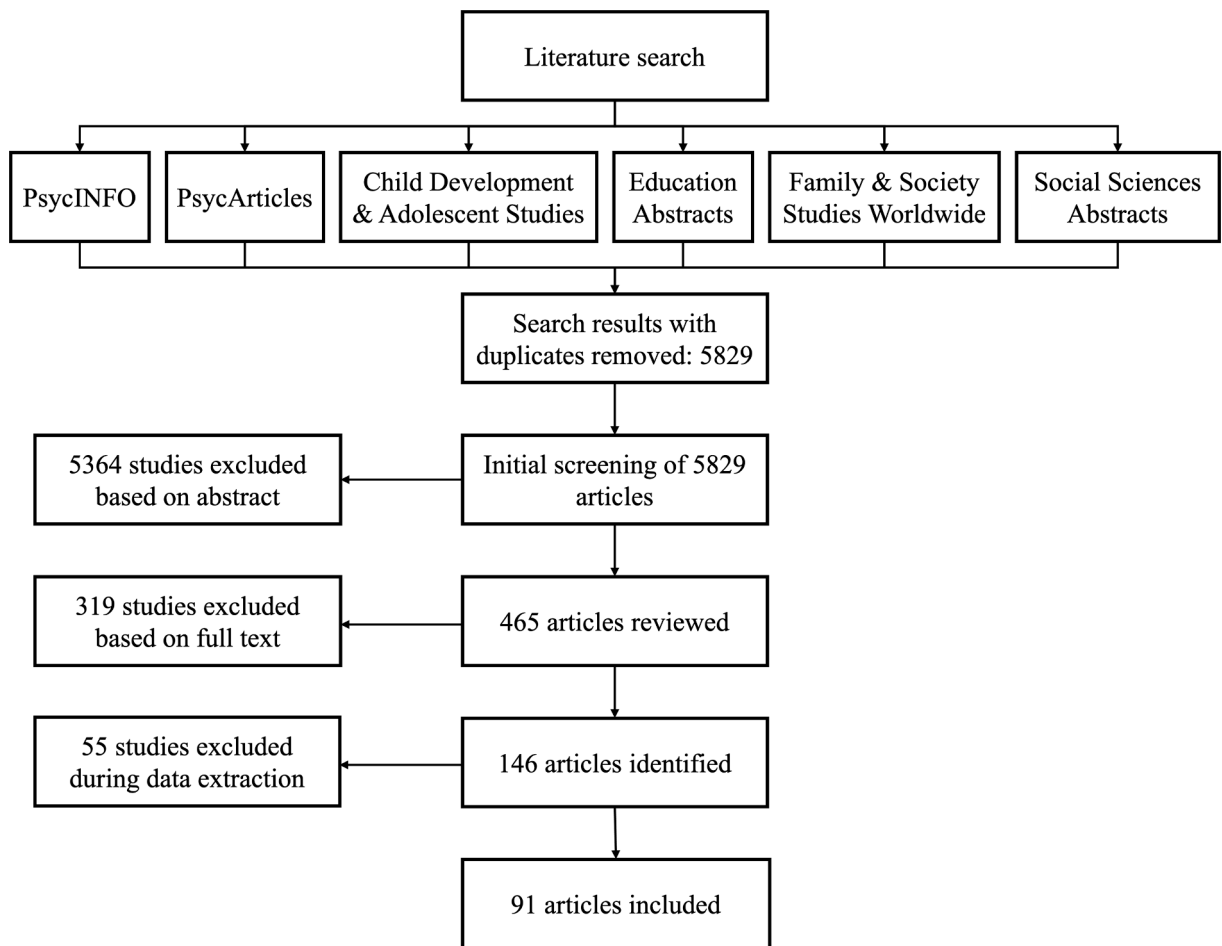


Fig. 1. Flow chart showing the number of studies at each step in the selection process.

conformed to a perfect Guttman scale, where if a child passed a later item, they passed all the prior items as well, skipping none.

Finally, studies were coded to indicate the country the children were from. We coded the specific country for each data point, but we additionally classified them into two large categories – individualistic and collectivist countries (Hofstede et al., 2010) – based on Hofstede's (2017) index of individualist values. Individualistic countries include the US., Canada, Poland, France, Germany, Portugal, Spain, Sweden, Switzerland, the UK, New Zealand, and Australia. Collectivist countries include China, Singapore, Indonesia, Iran, Pakistan, Turkey, Uganda, Vanuatu, Argentina, and Brazil.

All data analyses were conducted using the *metafor* package in R (Viechtbauer, 2010; R Core Team, 2023). Note that when multiple studies used overlapping samples, only the study with the largest sample was included in our data analyses. We first examined descriptive statistics for patterns and tendencies. We then used the Freeman-Tukey transformation (Freeman & Tukey, 1950) to transform the percentages and Guttman Coefficient of Reproducibility into a double arcsine measure that has better statistical properties (i.e., a sampling distribution that is closer to a normal distribution, with a more stable sampling variance). This double arcsine measure, coupled with the weighted matrices derived from the reciprocal of the sampling variances, was the focal variable that we used for performing all of our statistical analyses. In accordance with best practices for meta-analysis, we analyzed our data using random-effects models (with only one principled exception, as reported in our Results). After the data analyses, we did a corresponding back-transformation to derive the constitutive statistics that we display in reporting our results.

In our data analyses, we treated all data points as independent to simplify the statistical models, although five studies (Cavadel & Frye, 2017; Müller et al., 2012; Peterson & Wellman, 2019; Sarmento-Henrique et al., 2020; Song et al., 2016) were included that reported within-subjects data (e.g., they followed up the same child for two different time points). However, a subsequent analysis with a constrained dataset where every data point was independent confirmed the primary results. Additionally, there were seven unpublished data points from the “grey literature.” Since they exhibited the same general pattern as the published findings, they were thus incorporated into our main analyses.

Results

Descriptive statistics of study characteristics

Children's reported age averaged 5.03 years ($SD = 1.37$) and ranged from 2.48 to 10.2 years. The total number of children was 10,321 in the 141 data points, with an average of 73.19 children (median = 58) per data point. Among these 141 data points, 103 were collected in individualistic countries, and 38 were collected in collectivist countries. Fig. 2 provides a visualization for this breakdown. Although more data were collected in individualistic countries, data from collectivist countries were not uncommon, with a total of 4,506 (out of 10,321) children providing data from collectivist countries (mean age = 5.06 years; $SD = 0.92$; range = 3.32 to 8.25 years) compared to 5,815 (out of 10,321) children providing data from individualistic countries (mean age = 5.01 years; $SD = 1.51$; range = 2.48 to 10.2 years).

Overall passing percentage

Our focal research goal was to summarize the ToM Scale data across all available studies in the literature. We first calculated the overall percentage of passing each task (on average 122 data points contributed to this analysis across five tasks). Across five tasks and two cultural categories, we conducted ten meta-analyses using random-effects models to estimate the meta-analyzed overall passing percentages (tests of heterogeneity indicated significant heterogeneity across all models, with τ^2 ranging from 0.01 to 0.07, I^2 from 83 % to 94 %, and Q statistics from 222.44 to 1685.82; all $ps < 0.001$).

For individualistic countries, the results showed that the overall passing percentage for DD was 89.16 % (95 % confidence interval [86.75 %, 91.37 %]), 75.65 % for DB (95 % confidence interval [72.09 %, 79.04 %]), 64.81 % for KA (95 % confidence interval [58.97 %, 70.46 %]), 40.08 % for FB (95 % confidence interval [35.07 %, 46.65 %]), and 31.53 % for HE (95 % confidence interval [26.93 %, 36.31 %]). As summarized in Fig. 3A, the pattern of overall passing percentages follows the DD-DB-KA-FB-HE sequence found in Wellman and Liu (2004).

For collectivist countries, the results showed that the overall passing percentage for DD was 86.64 % (95 % confidence interval [83.76 %, 89.28 %]), 69.78 % for DB (95 % confidence interval [64.62 %, 74.70 %]), 72.05 % for KA (95 % confidence interval [67.29 %, 76.59 %]), 34.40 % for FB (95 % confidence interval [28.56 %, 40.48 %]), and 30.79 % for HE (95 % confidence interval [25.24 %, 36.63 %]). As shown in Fig. 3B, the pattern of overall passing percentages does not follow the original DD-DB-KA-FB-HE sequence; rather, it follows an alternative sequence, DD-KA-DB-FB-HE, where KA precedes DB.

Given that it is the KA and DB items that exhibit differences between the two cultural categories, we focused on them to examine whether there was an interaction effect between task and culture. A 2-by-2 moderation analysis (including the two tasks, DB and KA, and the two cultural categories, individualistic and collectivist countries) revealed a significant interaction effect between task and culture ($Z = 2.27$, $p < 0.05$; there were no main effects of task nor culture).² Children in individualistic countries passed DB before KA, while children in collectivist countries passed KA before DB.

To validate the robustness of these findings and address the issue of potential publication bias (where only significant findings are published or reported), we used funnel plots (visualized in Supplementary Figure S1) and conducted additional follow-up analyses using the Duval & Tweedie trim-and-fill method (Duval and Tweedie, 2000), as shown in Supplementary Figure S2, to estimate bias-corrected results. The bias-corrected results replicated and corroborated the previously reported general patterns: children from individualistic countries followed the DD-DB-KA-FB-HE sequence, while children from collectivist countries followed the DD-KA-DB-FB-HE sequence.

Guttman Coefficient of Reproducibility

In either case depicted in Fig. 3, in general, to evidence a Guttman scale, children's percent passing must not only reveal a sequential order, but a child has to pass all the preceding ones in order to pass a later step. Analysis of Guttman Coefficients of Reproducibility, a statistic that quantifies the extent to which a group of children adhere to the perfect Guttman sequence, confirmed that result (84 data points contributed to this analysis). Across two cultural categories, we conducted two meta-analyses using random-effects models (one for each cultural category); however, because in this case homogeneity was evident for both individualistic countries ($Q = 41.60$, $p = 0.95$) and collectivist countries ($Q = 22.90$, $p = 0.55$), the random-effects models were reduced to fixed-effect models to estimate the meta-analyzed Guttman Coefficient of Reproducibility.

For individualistic countries, the result showed that children followed the DD-DB-KA-FB-HE sequence; the Guttman Coefficient of Reproducibility was 0.95 (95 % confidence interval [0.94, 0.96]). For collectivist countries, the results showed that children followed the DD-KA-DB-FB-HE; the Guttman Coefficient of Reproducibility was 0.92 (95 % confidence interval [0.91, 0.93]). No other common sequences were found. As a value of 0.90 or greater is commonly considered significant for Guttman Coefficients of Reproducibility, both cases showed that children followed a consistent and scalable sequence in ToM development: DD-DB-KA-FB-HE for individualistic countries and DD-KA-DB-FB-HE for collectivist countries. The robustness of these findings was confirmed through funnel plots (see Supplementary Figure S3A) and Duval & Tweedie trim-and-fill analyses (Duval & Tweedie, 2000), with bias-corrected Guttman

² $\Delta\tau^2$ was 0.001 and R^2 was 4.18% after including task, culture, and their interaction as moderators.

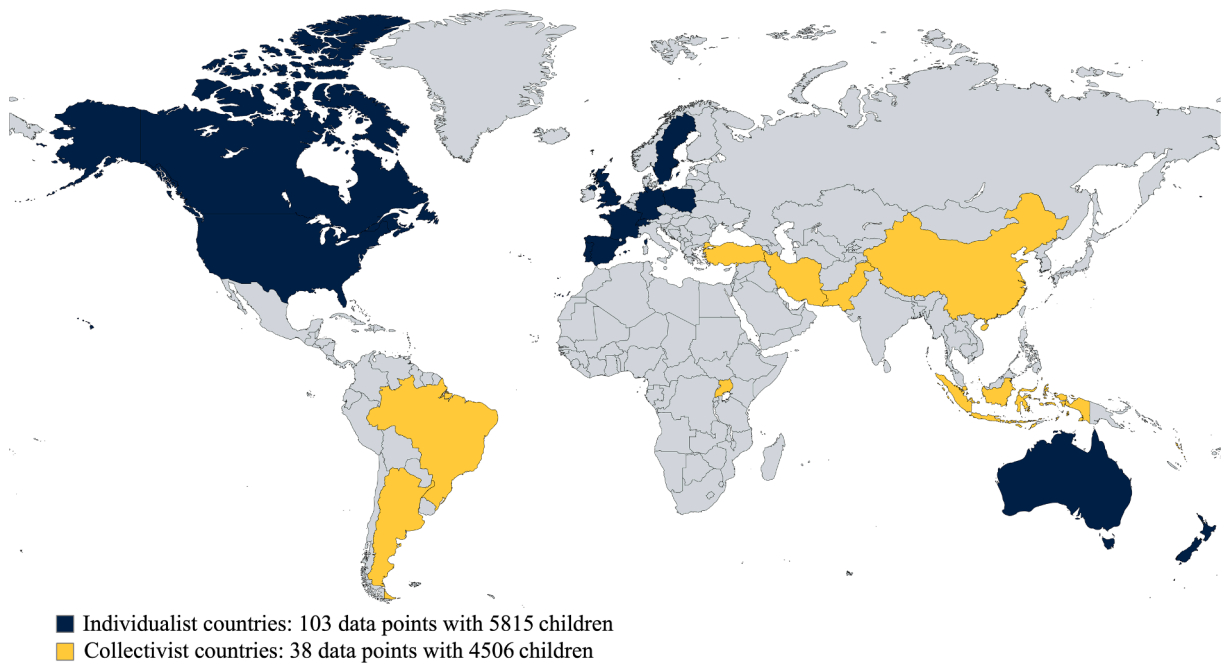


Fig. 2. Visualization of the individualistic and collectivist countries on a world map with the corresponding numbers of data points and sample sizes.

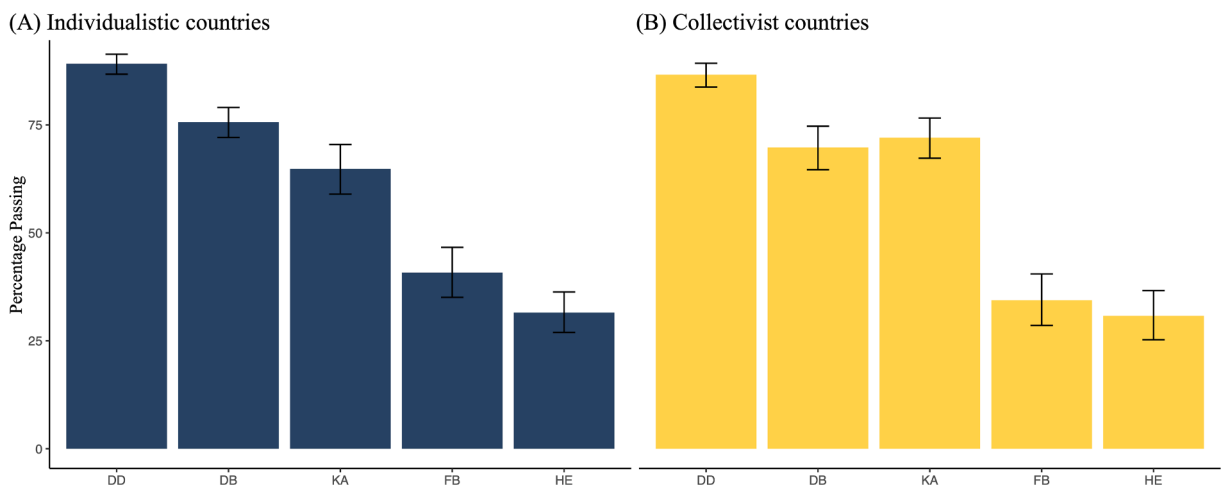


Fig. 3. The overall passing percentages for the five tasks in (A) individualistic countries and (B) collectivist countries.

Coefficients of 0.955 (95 % confidence interval [0.947, 0.962]) for individualistic countries and 0.92 (95 % confidence interval [0.91, 0.93]) for collectivist countries.

Additionally, we also looked at Green's index of Consistency. Green (1956) advocates for the use of this index of Consistency as well, which tests whether the observed Guttman Coefficient of Reproducibility was greater than chance (with a value of 0.50 or greater considered significant; 71 data points contributed to this analysis). As done for the analysis of Reproducibility, we conducted two meta-analyses across two cultural categories (one for each cultural category) using random-effects models (tests of heterogeneity revealed significant heterogeneity in both models, with $\tau^2 = 0.018$ and 0.017 , $I^2 = 81\%$ and 89% , and Q statistics of 246.04 and 222.59 for individualistic and collectivist countries, respectively; both $ps < 0.001$). For these studies Green's index of Consistency was 0.73 (95 % confidence interval [0.69, 0.77]) for children from individualistic countries and it was 0.63 (95 % confidence interval [0.58, 0.69]) for children from collectivist countries, again confirming the findings reported above, both greatly exceeding the 0.50 level needed for significance. These findings were further confirmed through funnel plots (see [Supplementary Figure S3B](#)) and Duval & Tweedie trim-and-fill analyses (Duval & Tweedie, 2000), with bias-corrected Green's indexes of 0.73 (95 % confidence interval [0.69, 0.77]) for

individualistic countries and 0.64 (95 % confidence interval [0.68, 0.69]) for collectivist countries.

Age as a Factor

As ToM develops during the preschool period, the performance on ToM Scale items can systematically change with age. As shown above, the timeline for this development can also differ for children growing up in different cultures and countries, with children achieving correct performance on a particular ToM milestone at somewhat earlier or later ages, depending on where they are raised. Because the children tested differed in age across studies, our data allowed us to examine this developmental profile in addition to the overall results reported so far.

We began with the analysis of ToM Scale total scores (0 to 5, depending on how many items a child passed). We converted the raw total scores to range from 0 to 100, conceptually equivalent to the percentage of passing the entire Scale and mathematically equivalent to the average passing percentage across the five tasks. Then two meta-regressions were conducted using random-effects models (one for each cultural category). As depicted in Fig. 4, results, for children from both individualistic and collectivist countries, showed an upward trajectory with increasing ages. Meta-regression analyses revealed a significant main effect for age in individualistic countries ($Z = 10.88, p < 0.001$).³ In collectivist countries, the smaller sample size and narrower age range attenuated this trend, but the positive slope was still apparent and resulted in a marginally significant main effect for age ($Z = 1.93, p = 0.05$).⁴ Thus, as expected, ToM understandings indexed by the ToM Scale show a clear trajectory of ToM development with increasing age.

This pattern is further confirmed by ten item-by-item meta-regressions across the five tasks and two cultural categories using random-effects models as depicted in Fig. 5. The effects of increasing age were consistent across the five tasks, with meta-regression analyses yielding significant main effects for age in individualistic countries (all Z s > 6.31 and all p s < 0.001 across five tasks).⁵ In collectivist countries, again, the smaller sample size introduced more variance attenuating the regressions, but the trend of positive slopes was still evident (p s ranging from .01 to .5).⁶

Note that in both cultural contexts, understanding of DD approached ceiling (100 % correct) early in development. In contrast, understanding of HE was at floor (0 % correct) early in development, improved with age, and often failed to reach consistently correct performance even in older children (7 and 8 + years), again, in both cultural contexts. Performance on the other items fell in between in sequential order.

Using the results reported in Fig. 5, we further computed the estimated age when 70 % of children passed a task. We used 70 % as an arbitrary but intuitively revealing threshold. As shown in Fig. 6A, in individualistic countries even children younger than 2.48 years were, on average, 70 % correct on DD; children were passing DB on average at 4.18 years; at 5.40 years for KA; at 7.10 years for FB; and only at an average of 9.37 years for HE. Note that because the minimum age reported in all the studies of individualistic countries was 2.48 years, the age estimate of passing DD is truncated. But, clearly, many 2-year-olds understand that people can have diverse desires (i.e., DD).

For collectivist countries, as shown in Fig. 6B, the results demonstrated that even children younger than 3.32 years were 70 % correct on DD; 70 % of children were passing KA at average at 3.99 years, but were 70 % on DB only at 5.20 years; and 70 % were passing FB and HE only at 8.25 + years. Again, because the minimum age reported in all the studies of collectivist countries was 3.32 years, the age passing DD was truncated, but it was certainly less than 3.32. Similarly, because the maximum age was 8.56 years, we could not provide precise estimates for FB and HB as well. In this case, although the results on FB and HE look the same for 70 % of children passing, FB indeed developed earlier than HE. For example, at the maximum age of 8.25 years, 59.2 % of children could pass FB, but only 46.3 % of children could also pass HE.

In sum, these results, summarized in Fig. 6, were consistent with the conclusion that ToM universally develops in a progressive sequence of understandings, and that children from individualistic and collectivist countries evidence two slightly different, culturally specific sequences.

Longitudinal data

The ToM Scale reveals sequences of development via a cross-sectional analysis of the orders of difficulty of the target items across numerous children as reported above. By hypothesis, this order also reflects the sequence of understanding individual children go through as they grow older. However, that can only be directly tested using longitudinal data. Fortunately, although rare, eight studies tested children on the Scale at least twice over intervals of months or years (Cavadel & Frye, 2017; Ding et al., 2015; Müller et al., 2012; Peterson & Wellman, 2019; Rhodes & Wellman, 2013; Sarmento-Henrique et al., 2020; Song et al., 2016; Wellman et al., 2011). These eight studies include children from individualistic countries (Australia, Spain, and the US) but also collectivist ones (China). Although not all of these studies contributed multiple (longitudinal) data points to our quantitative meta-analysis (only five did so), these studies qualitatively confirmed that individual children pass through the tasks in order as they grow older, just as reflected in the cross-sectional data.

³ Individualistic countries: $\Delta\tau^2$ was 0.018 and R^2 was 76.96% after including age as a moderator.

⁴ Collectivist countries: $\Delta\tau^2$ was 0.01 and R^2 was 10.52 % after including age as a moderator.

⁵ Individualistic countries: For DD, DB, KA, FB, and HE, respectively, $\Delta\tau^2$ were 0.008, 0.014, 0.033, 0.044, and 0.031, and R^2 were 39.65%, 49.05%, 57.98%, 63.49%, and 51.73% after including age as a moderator.

⁶ Collectivist countries: For DD, DB, KA, FB, and HE, p s range from .01 to .5; all $\Delta\tau^2$ and R^2 were not significant.

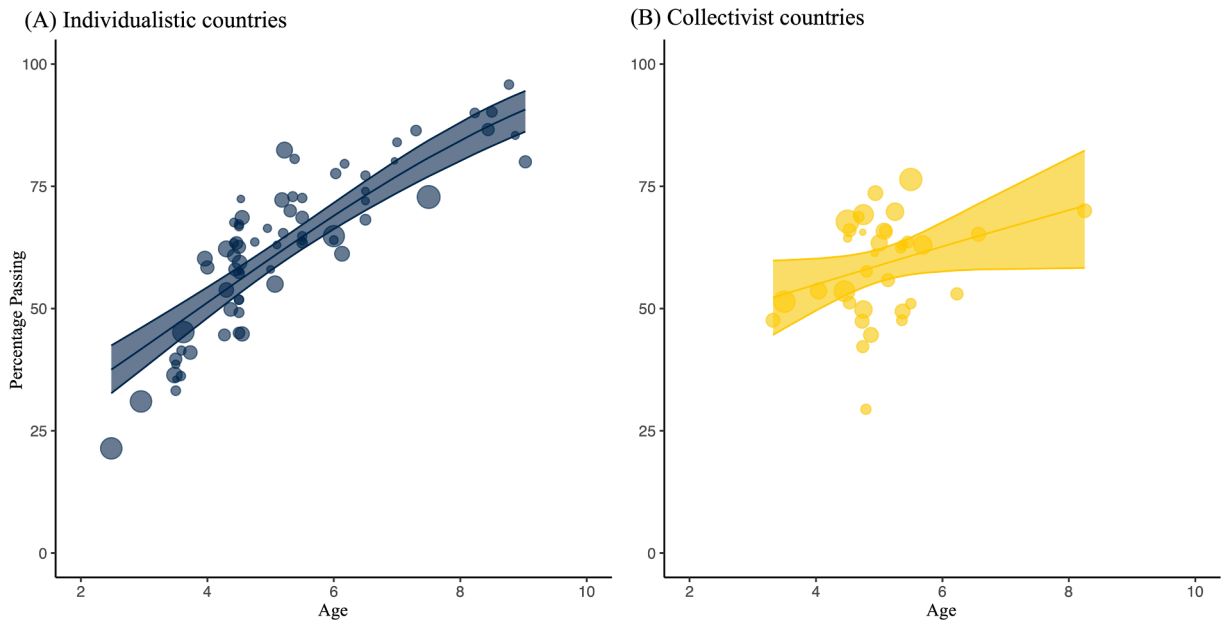


Fig. 4. The effect of age for the ToM Scale total scores across two cultural categories. Each circle indicates one data point, the larger size of the circle, the larger sample size of the underlying sample. The x-axis shows the reported average age for each data point. The y-axis shows the passing percentage of the Scale (converted to percentages from total scores). The best fitting regression lines are also shown, with the shaded areas indicating the 95 % confidence intervals for those best-fitting regression lines.

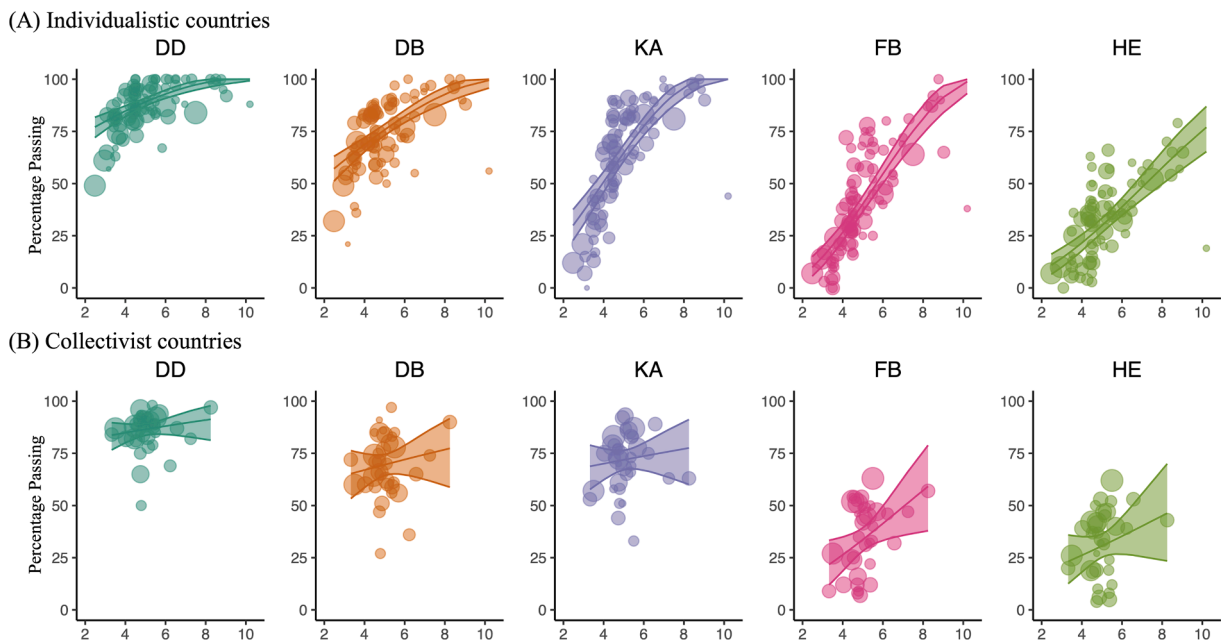


Fig. 5. The effect of age across five tasks and two cultural categories. Each circle indicates one data point, the larger size of the circle, the larger sample size of the underlying sample. The x-axis shows the reported average age for each data point. The y-axis shows the reported average percentage correct for the children encompassed by each data point. The best fitting regression lines are also shown, with the shaded areas indicating the 95 % confidence intervals for those best-fitting regression lines.

Discussion

In this meta-analysis, we aimed to synthesize the abundant data from studies using the 2004 Wellman and Liu Theory of Mind (ToM) Scale in order to describe how children worldwide progress through a sequence of ToM understandings in their developing

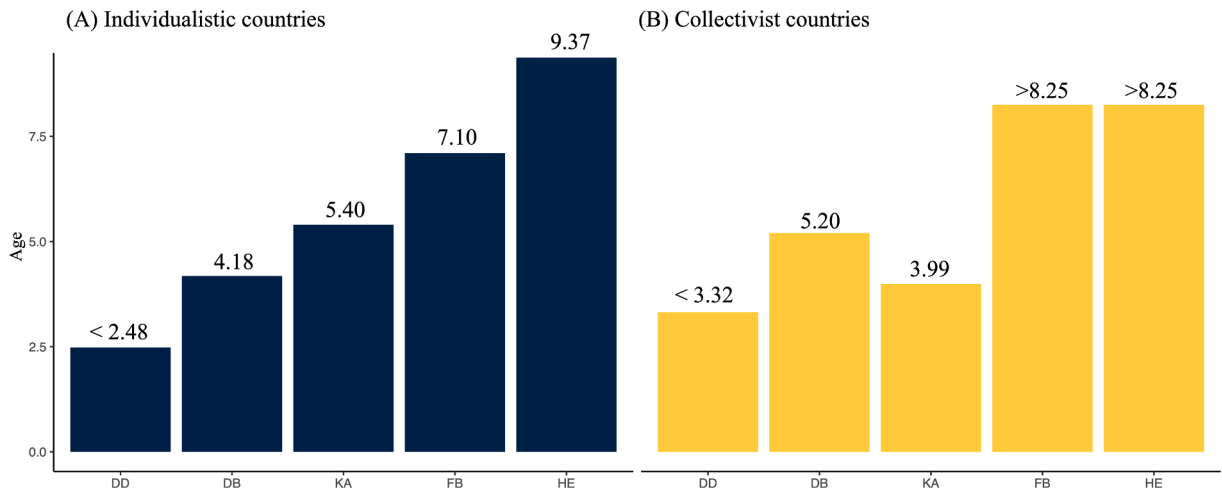


Fig. 6. The estimated age of passing the five tasks (when 70% of the children could pass) in (A) individualistic countries and (B) collectivist countries.

insights about people and minds. Despite its widespread use, patterns of findings using the ToM Scale had only previously received partial, narrative reviews. Our meta-analysis has now aggregated arguably all the extant, available findings (as of 2024) to provide a comprehensive summary of ToM Scale results. The meta-analytic results show that the ToM Scale and its five tasks, from DD, DB, KA, FB, and HE, successfully capture a robust, widespread age-related, developmental progression of ToM understandings and captures such developments in a culturally sensitive manner.

In our introduction, we outlined the potential advantages of meta-analyses in general and for this topic in particular: Meta-analyses allow researchers to address issues of replicability and failure to replicate, a much-discussed topic in contemporary psychological science. They also help address problems that accrue with small-sample, underpowered research studies. Additionally, they may help overcome the WEIRD (western, educated, industrialized, rich, and democratic; [Henrich et al., 2010](#)) sample bias in research by including data from less represented populations. Our meta-analysis evidences all these advantages with regard to ToM development leading to several clear conclusions as highlighted below.

First, it is now clear that many studies worldwide evidence the same, replicable sequence of ToM development. The aggregated data showed the sequences arise again and again, essentially unanimously, across all the constituent, smaller-scale studies. The sequences showed clear patterns of scalability, where it was rare that a child passed a later item without passing all the earlier items as well. This finding was systematically confirmed by significant Guttman's Coefficients of Reproducibility and Green's index of Consistency ([Green, 1956](#)). While many of the individual studies might have been underpowered, the aggregate data certainly were not – they represent close to 11,000 children worldwide who came almost equally from non-WEIRD as well as WEIRD societal contexts.

Substantively, our findings demonstrated that typically-developing children understood that people's desires could differ (i.e., DD) before age 2 to 3 years. Then, they understood that persons could have differing beliefs (i.e., DB) and could be knowledgeable or ignorant (i.e., KA). After that, by the end of the preschool years, about 7+ years, they understood that people could have false beliefs (i.e., FB), an often-cited milestone of ToM competence. It was only after understanding FB that children came to know that people could feel something inside but display an altogether different emotion on their face (i.e., HE), a decidedly post-preschool insight.

Thus, as predicted, in terms of timetables, most typically-developing children, across the many countries and contexts represented in our dataset have developed all or most of these understandings during the preschool years. As shown in [Figs. 5 and 6](#), indeed, by age 7, most children have developed one of the most complex ToM understandings, false-belief understanding (i.e., FB), as well as all the preceding ones.

Furthermore, as predicted, in terms of universality and specificity, both are evident. On the one hand, cultural specificity was clear because there were two culturally specific sequences: Children from individualistic countries understood that people's beliefs could differ (i.e., DB) early on and before understanding that people can be knowledgeable or ignorant (i.e., KA) in a DD-DB-KA-FB-HE sequence. Children from collectivist countries understood basic aspects of knowledge acquisition (as indexed by KA) in advance of understanding about diversity of beliefs (DB) in a DD-KA-DB-FB-HE sequence. On the other hand, universality was clear because the two sequences were largely identical, and no other common sequences arose. In short, there was constrained variability within a highly common sequence of preschool acquisitions.

Notably, the sequences of development revealed via this cross-sectional ToM Scale also represent the longitudinal steps that individuals pass through as they grow older. This has been confirmed in a set of eight longitudinal studies where the same children were tested on the ToM Scale two or three times over intervals of months or years, with samples from both individualistic countries and collectivist ones ([Cavadel & Frye, 2017](#); [Ding et al., 2015](#); [Müller et al., 2012](#); [Peterson & Wellman, 2019](#); [Rhodes & Wellman, 2013](#); [Sarmiento-Henrique et al., 2020](#); [Song et al., 2016](#); [Wellman et al., 2011](#)).

The differences in the two dominant sequences deserve additional explanations. As has been argued elsewhere, a plausible

explanation arises from the cultural and child-raising differences apparent in individualistic as compared to collectivist contexts which can be outlined more specifically by considering the US versus China (Li, 2002; Nisbett, 2004). Back to Confucius, Chinese culture has long prized attaining the common knowledge that all right-minded people should know. Individualist cultures put greater weight on understanding beliefs, including emphasizing not only the truth or falsity of beliefs, but belief divergence: Individuals may well have, and indeed have the right to, their own beliefs. These differing cultural priorities are apparent in adult conversations with young children as well. For example, parents in China talk to their children frequently about knowledge using a variety of terms (especially verbs) for knowledge and knowing (e.g., Tardif & Wellman, 2000). Parents in the U.S. more often talk about beliefs, thinking, thoughts, and differences in beliefs (e.g., Ruffman et al., 2002). Furthermore, empirical evidence shows that Australian and US parents encourage their children to assert their own opinions and acknowledge persons' differing views (e.g., Sigel et al., 1992), whereas Chinese parents discourage the voicing of discordant personal views while emphasizing culturally agreed-upon ideas and know-how (e.g., Johnston & Wong, 2002). These well-documented cultural differences sensibly explain the relatively early understanding of knowledge as indexed by KA and a KA-DB sequence in one case, as compared to the relatively early understanding of beliefs and differences in beliefs (DB) and a DB-KA sequence in the other.

These findings and related theorizing speak to several theoretical alternatives in ToM development. By hypothesis, a sequence could result, on the one hand, from innately programmed modular maturations (e.g., Leslie et al., 2004) or, on the other hand, from processes of experience-dependent constructivist learning in which initial conceptions lead to later concepts shaped by relevant information and experience (Gopnik & Wellman, 2012). Overall, our data favor a non-nativist learning position where ToM development results from experience-dependent learning informed by culturally specific parental- and community-provided information and practices.

It is important to note that, given the consistent scalability of our results, even the ToM Scale total scores (0 to 5) reported in past or future studies provide useful sequential data unavailable in other batteries of tasks. A total score of 3 on this ToM Scale almost always means that DD, DB, and KA tasks were passed, not just any 3 of the 5 tasks. Also, this property means that if a study uses just one task, such as the much-used false belief task, it can arguably assert that children who have passed that task have passed earlier ones as well. This scale-related understanding of false belief scores helps clarify and extend other important ToM findings. For example, variation in understanding false belief has been the predictor in numerous studies showing that ToM developments have wide-ranging consequences for children's everyday lives, including significant impacts on children's friendships and popularity, engagement in lying and deception, game-playing skills, strategies for persuading or arguing with others, and transition to school (e.g., Lecce et al., 2017; Peskin & Ardino, 2003; Slaughter et al., 2015; Talwar & Lee, 2008). Our data suggest that the development of a cumulative sequence of ToM understandings, rather than merely a single false belief milestone, may underwrite such relationships.

Limitations and future Directions

Despite its many strengths, several limitations of the current study deserve mention. These limitations also set the stage for needed future research.

Importantly, the studies encompassed by this meta-analysis come from 22 countries distributed across the globe (see Fig. 2). However, more than 190 countries are represented in the United Nations. Thus, in that broader context, considering only 22 limits a detailed assessment of the global universality and specificity of ToM development. Moreover, individualistic contexts (contributing 104 data points) were overweighted in our data compared to collectivist ones (contributing 39 data points, where several countries contributed only a single data point). This disparity resulted in a restricted age range and larger variance for collectivist countries, yielding less precise developmental estimations. Fortunately, the ToM Scale has proven easy to translate into many contexts using community-specific wordings and culturally-appropriate figurines, toys, and drawings. And, because it includes only five short tasks it could easily be incorporated into further studies in further locales.

Our use of an individualistic versus collectivist contrast via a broadly utilized classification scheme (Hofstede et al., 2010) provides informative results. Still, it notably oversimplifies a complex cultural landscape into binary categories (e.g., Kryś et al., 2022). This dichotomy undoubtedly fails to acknowledge other cultural dimensions as well as the nuanced variance within individual countries. For instance, many countries in South America, often regarded as collectivist (i.e., Argentina and Brazil via data and classification in Hofstede et al., 2017), display varying degrees of cultural values within different regions. Recent studies in Brazil, for example, have shown that individuals living in different parts of Brazil demonstrate different emphases on individualist or collectivist values, challenging the notion of homogenous cultural classifications. These limitations underscore the need for more extensive, large-scale cross-cultural research endeavors, especially in a more comprehensive array of underrepresented cultures. Moreover, more research that includes measures of specific cultural values is needed. For instance, it would be important to know the cultural values and practices in the families of the children whose ToM is being tested—something that was notably lacking in almost all studies in the current meta-analysis. This could be accomplished by observing parent-child family practices or asking parents about them. Such investigations would substantially enhance our understanding of the interplay between ToM development and cultural factors, providing more definitive evidence regarding the universal and culture-specific aspects of ToM development.

Plausibly, because children understand individual differences in desires early on, this prompts them to consider and look for other mind-mind and mind-world differences, including that beliefs can differ between persons. That is, in a Bayesian fashion, where prior understandings help shape new data observations leading to revised posterior understanding, it may well be that early childhood understanding of diverse desires (i.e., DD) forms the prior understanding for amassing further evidence that leads to a posterior understanding of differences in belief (i.e., DB), which forms a prior for learning about differences in knowledge and ignorance (i.e., KA), and so on. However, sequential order does not necessarily indicate causality. There may well be other equally or even more powerfully

formative understandings at work. For example, children's increasing understanding of emotions, moral and immoral actions, or personality differences (e.g., shy versus outgoing participants) may well power ToM developments in the preschool years. More experimental and/or training research using the ToM Scale (e.g., Ding et al., 2015; Rhodes & Wellman, 2013) is needed to understand how sequential understandings unfold causally to create ToM development.

Conclusions

The present meta-analysis summarizes and describes the ToM Scale data from the past 20 years and offers new insights into our understanding of how theory of mind develops, as well as its universality and cultural specificity. In advance of and setting the stage for future research, our data contribute to a more comprehensive account of ToM development, and indeed social cognitive development more generally, by showing the progressive patterns, similarities, and differences in sequential ToM understandings that characterize child social-cognitive development worldwide.

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CRediT authorship contribution statement

Chi-Lin Yu: Data curation, Formal analysis, Investigation, Software, Conceptualization, Methodology, Validation, Project administration, Funding acquisition, Writing – original draft, Writing – review & editing. **Henry M. Wellman:** Supervision, Conceptualization, Methodology, Validation, Project administration, Funding acquisition, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data for the meta-analysis and the associated analysis script is available online (<https://doi.org/10.7302/e0jx-7g04>).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.dr.2024.101162>.

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